

DECISION TREE: (FOR CLASSIFICATION)

• Supervised learning algorithm

• Reg → continuous (IV → num; DV → num)

Class → - - -

classification regression

• Recursively splitting the dataset into subsets on the best feature that maximizes purity (classification) or minimizes error (regression).

BEST FEATURE

IMPURITY MEASURE

• metrics used to evaluate the HOMOGENEITY (or HETEROGENEITY) in dataset.

MATHEMATICAL FOUNDATION

A DT select splits using impurity measures.

(A) ENTROPY:

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \cdot \log_2(p_i)$$

p = proportion

c = no. of class

if Ent(S) = 0

Pure node

if Ent(S) = 1

Max. impurity

$$\log_2(n) = \frac{\log(n)}{\log(2)}$$

eg: 5 boys, 9 girls

$$\rightarrow - \left[\frac{5}{14} \log_2\left(\frac{5}{14}\right) + \frac{9}{14} \log_2\left(\frac{9}{14}\right) \right]$$

(B) INFORMATION GAIN

$$IG(S,A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

A = Feature

S_v = subset after split

(C) GINI INDEX

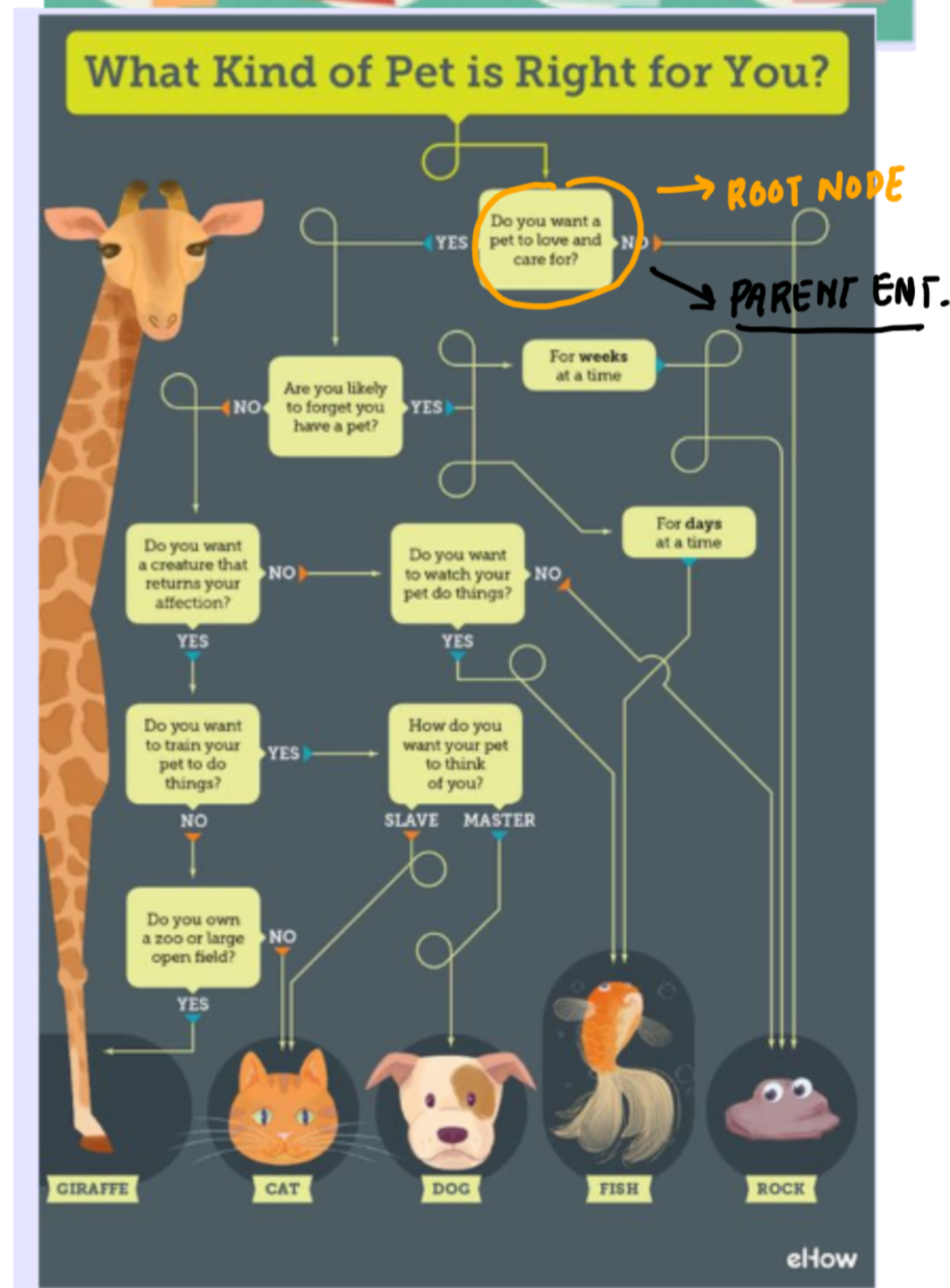
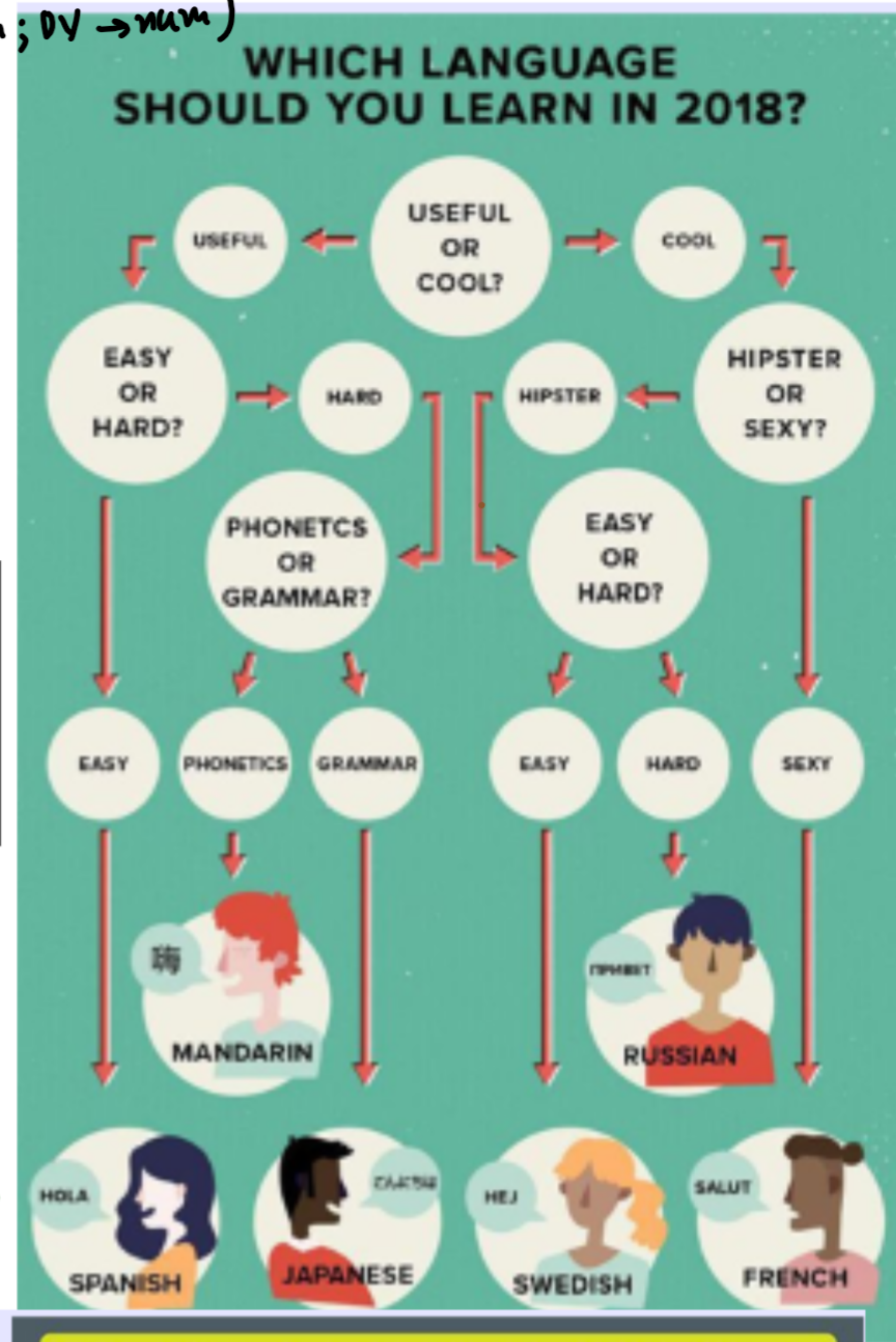
$$\text{Gini}(S) = 1 - \sum_{i=1}^c p_i^2$$

Gini(S) = 0

Pure

Gini(S) = 0.5

Max. impurity.



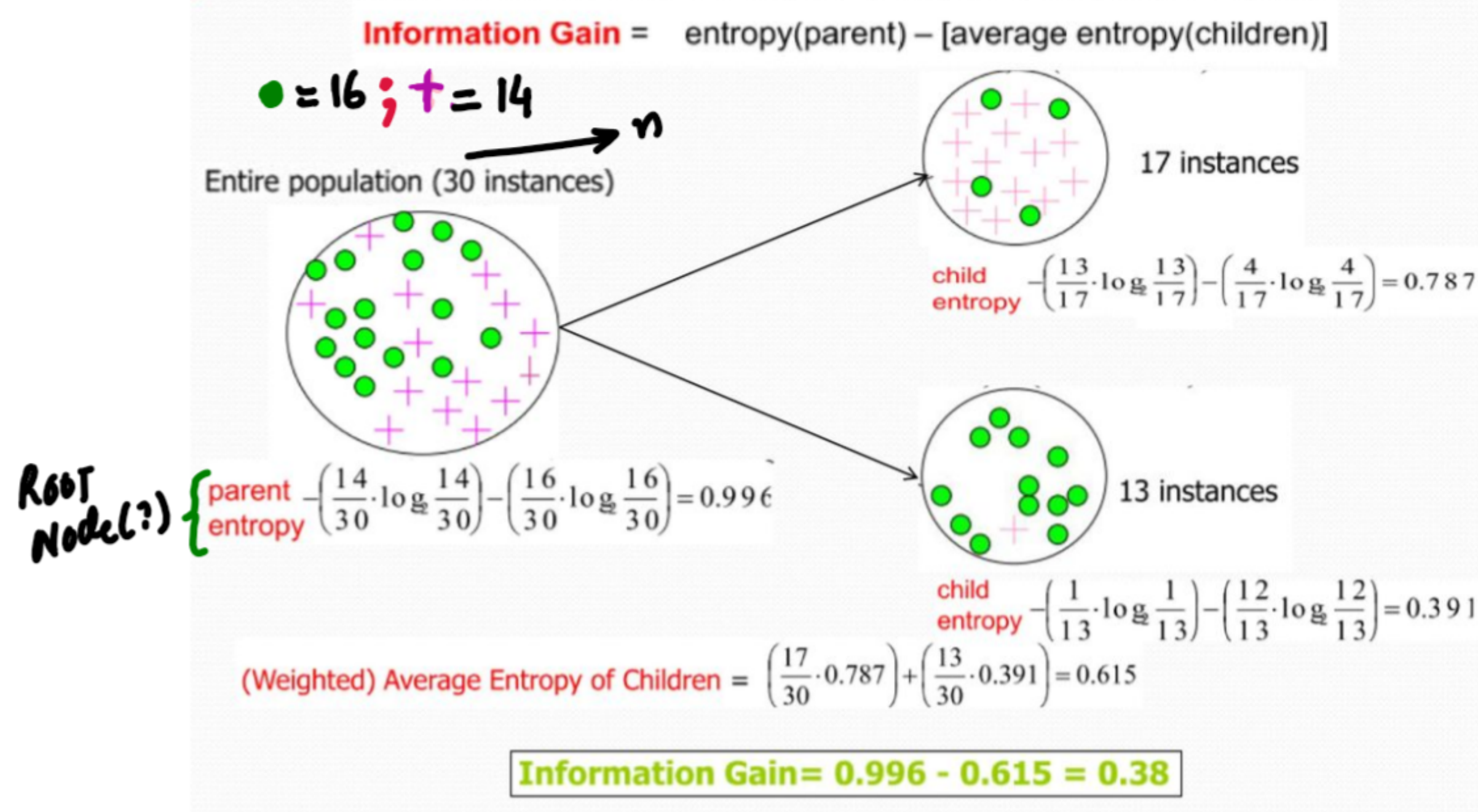
Exp: 1

Consider the following data, where the Y label is whether or not the child goes out to play.

Day	F_1 Weather	F_2 Temperature	F_3 Humidity	F_4 Wind	Y Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Exp: 2

Information gain calculation example



STEP-I: Entropy (S) $\rightarrow$ related to output (y)
 $n=10$
 $Y_{ES}=5 ; N_{O}=5$
 y is classification

$$P(YES) = P_1 = \frac{5}{10}$$

$$P(NO) = P_2 = \frac{5}{10}$$

$$\begin{aligned} \text{Entropy}(S) &= - \left[\frac{5}{10} \log_2 \left(\frac{5}{10} \right) + \frac{5}{10} \log_2 \left(\frac{5}{10} \right) \right] \\ &= - \left[(0.5 \times -1) + (0.5 \times -1) \right] \\ &= - \left[-0.5 - 0.5 \right] \end{aligned}$$

$$\boxed{\text{Entropy}(S) = 1}$$

for ref: $\log_2 \left(\frac{5}{10} \right) = \frac{\log_2 \left(\frac{5}{10} \right)}{\log_2 (2)} = -1$

STEP-II: INFORMATION GAIN FOR FEATURE-I (F_1)

F_1 WEATHER

$n=10$

S_v Sunny=3; Cloudy=3; Rainy=4

$Y=1, N=2$ $Y=3, N=0$ $Y=1, N=3$

$$\textcircled{1} - \left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3} \right) = 0.917$$

$$\textcircled{2} - \left(\frac{3}{3} \log_2 \frac{3}{3} + \frac{0}{3} \log_2 \frac{0}{3} \right) = 0$$

$$\textcircled{3} - \left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4} \right) = 0.81$$

$$Y=1, N=2$$

$$P_1 = \frac{1}{3} ; P_2 = \frac{2}{3}$$

$$Y=3, N=0$$

$$P_1 = 1 ; P_2 = 0$$

$$Y=1, N=3$$

$$P_1 = \frac{1}{4} ; P_2 = \frac{3}{4}$$

$$\text{Ent}(S) = \sum \frac{|S_v|}{|S|} \cdot \text{Ent}(S)$$

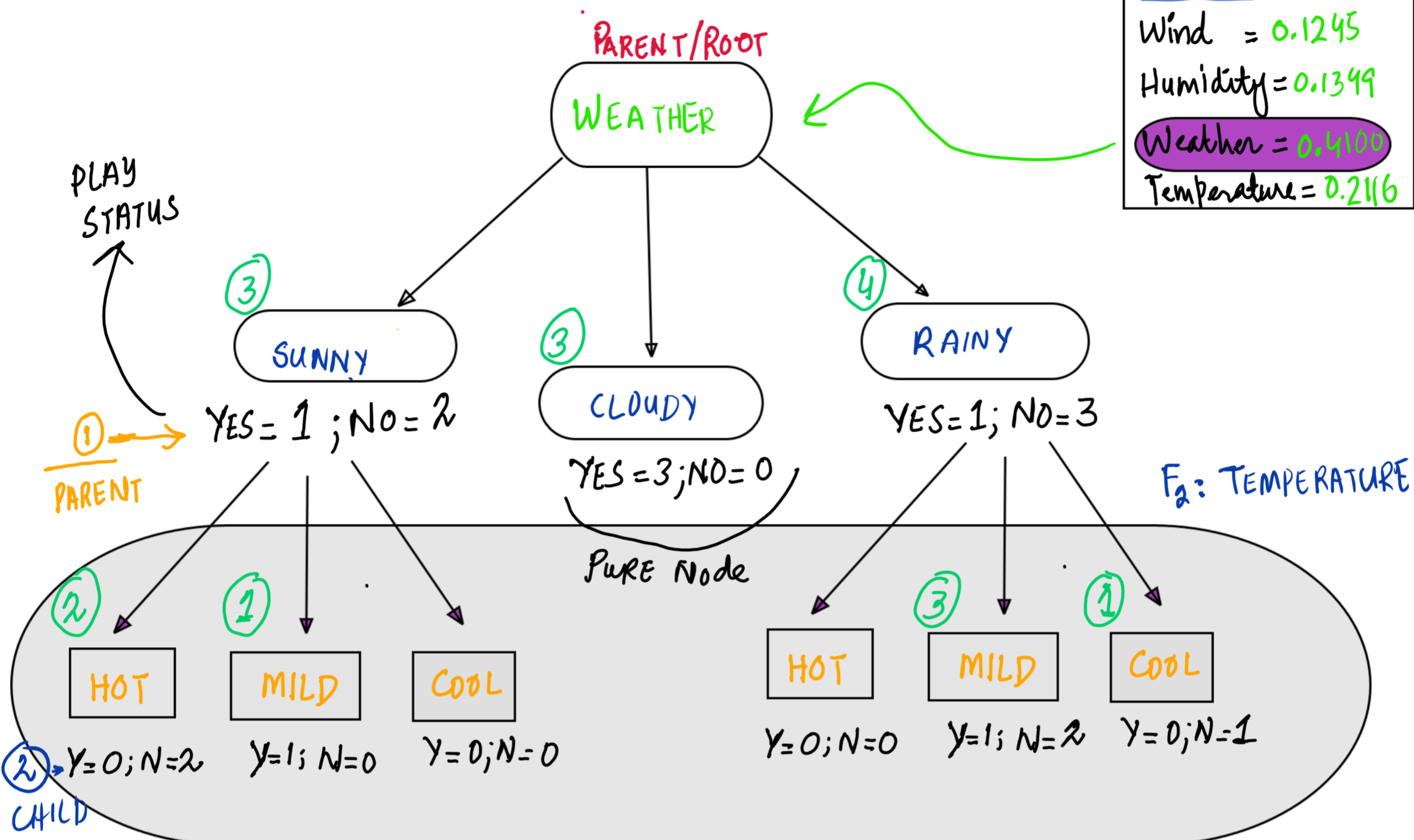
$$\begin{aligned} IG &= 1 - \left[\frac{3}{10} (0.917) + \frac{3}{10} (0) + \frac{4}{10} (0.81) \right] \cdot 1 \\ &= 1 - 0.59 \\ IG &= 0.41 \end{aligned}$$

TASK=

Find IG for F_2, F_3, F_4 !

Compare which feature has more IG among others?

IG status:
 Wind = 0.1245
 Humidity = 0.1349
Weather = 0.4100
 Temperature = 0.2116



①
 $\rightarrow Ent(Sunny) = -\left(\frac{1}{3} \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \log_2\left(\frac{2}{3}\right)\right)$
 $= 0.9183$

②
 $Ent(temp) = -\left[\frac{2}{3} \left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{2}{2} \log_2 \frac{2}{2}\right) + \frac{1}{3} \left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1}\right) + 0\right]$
 $= 0$

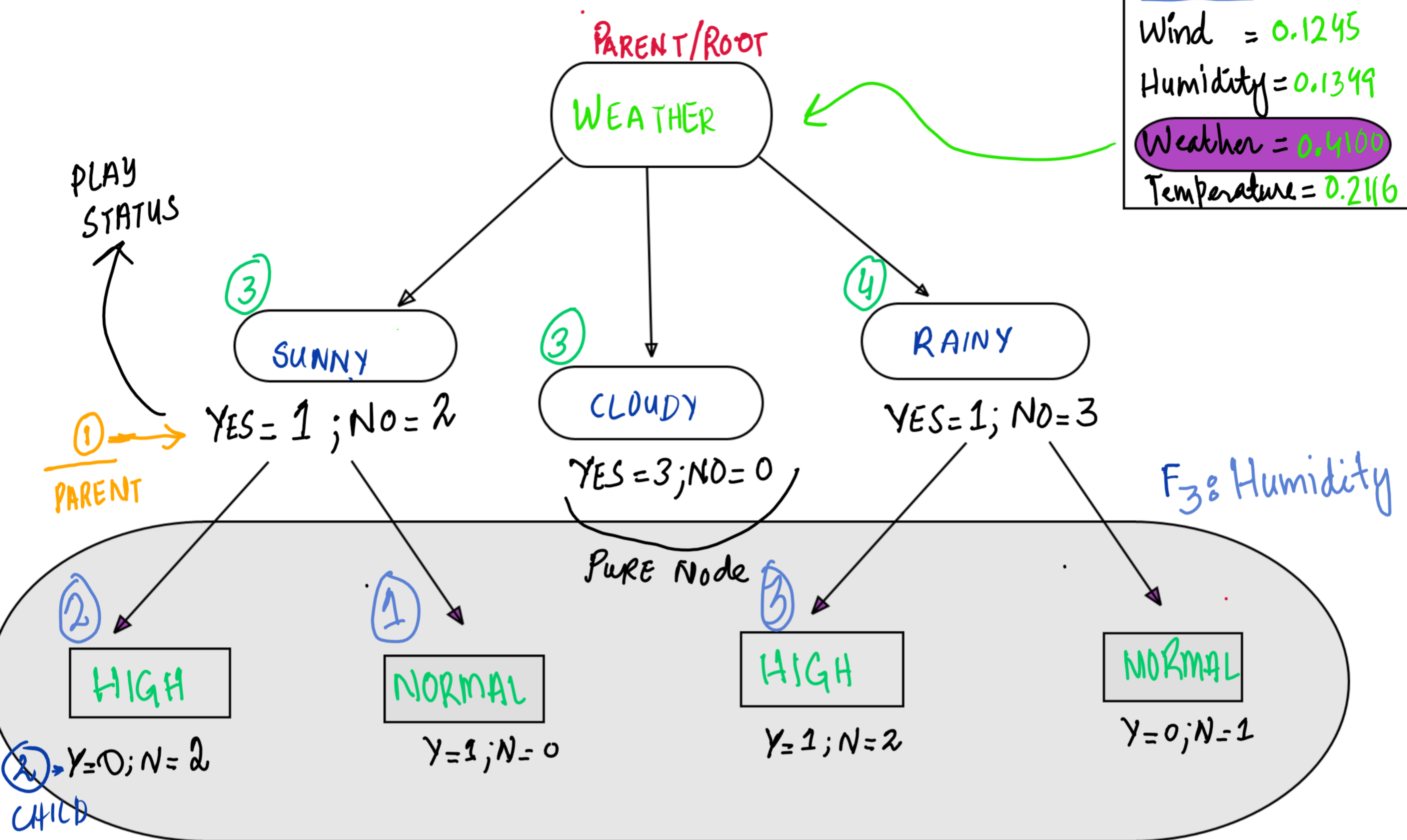
IG = 0.9183 - 0 = 0.9183

①
 $\rightarrow Ent(Rainy) = -\left(\frac{1}{4} \log_2\left(\frac{1}{4}\right) + \frac{3}{4} \log_2\left(\frac{3}{4}\right)\right)$
 $= 0.8113$

②
 $Ent(temp) = -\left[0 + \frac{3}{4} \left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3}\right) + \frac{1}{4} \left(\frac{0}{0} \log_2 \frac{0}{0} + \frac{1}{1} \log_2 \frac{1}{1}\right)\right]$
 $= 0.6887$

IG = 0.8113 - 0.6887 = 0.1226

IG status:
 Wind = 0.1245
 Humidity = 0.1349
Weather = 0.4100
 Temperature = 0.2116



①

$$\text{Ent}(\text{Sunny}) = - \left(\frac{1}{3} \log_2 \left(\frac{1}{3} \right) + \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \right)$$

$$= 0.9183$$

②

$$\text{Ent}(\text{Hum}) = - \left[\frac{2}{3} \left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{2}{2} \log_2 \frac{2}{2} \right) + \frac{1}{3} \left(\frac{1}{1} \log_2 \frac{1}{1} + \frac{0}{1} \log_2 \frac{0}{1} \right) \right]$$

$$= 0$$

IG = 0.9183 - 0 = 0.9183

①

$$\text{Ent}(\text{Rainy}) = - \left(\frac{1}{4} \log_2 \left(\frac{1}{4} \right) + \frac{3}{4} \log_2 \left(\frac{3}{4} \right) \right)$$

$$= 0.8113$$

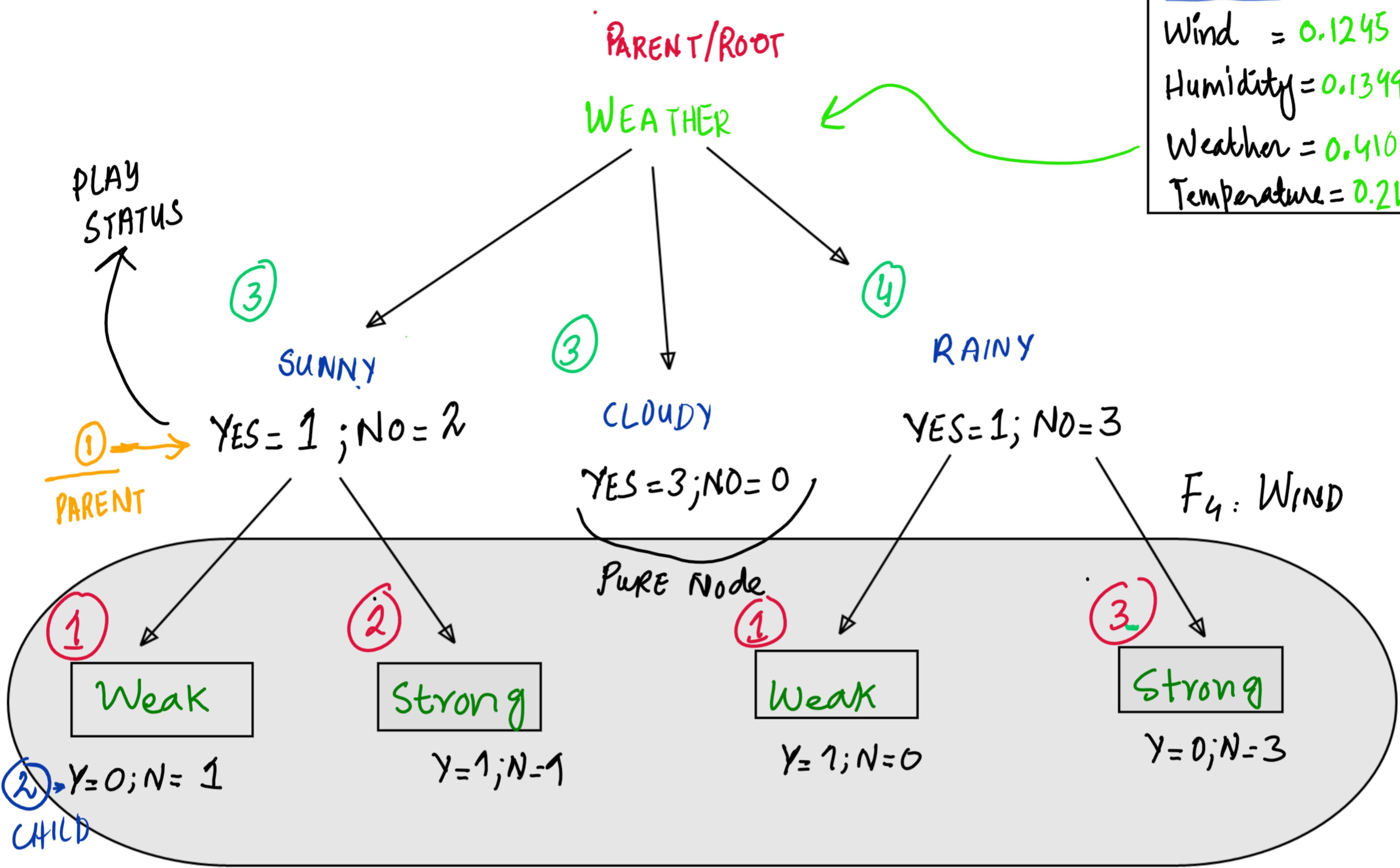
②

$$\text{Ent}(\text{temp}) = - \left[\frac{3}{4} \left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3} \right) + \frac{1}{4} \left(\frac{0}{0} \log_2 \frac{0}{0} + \frac{1}{2} \log_2 \frac{1}{2} \right) \right]$$

$$= 0.6887$$

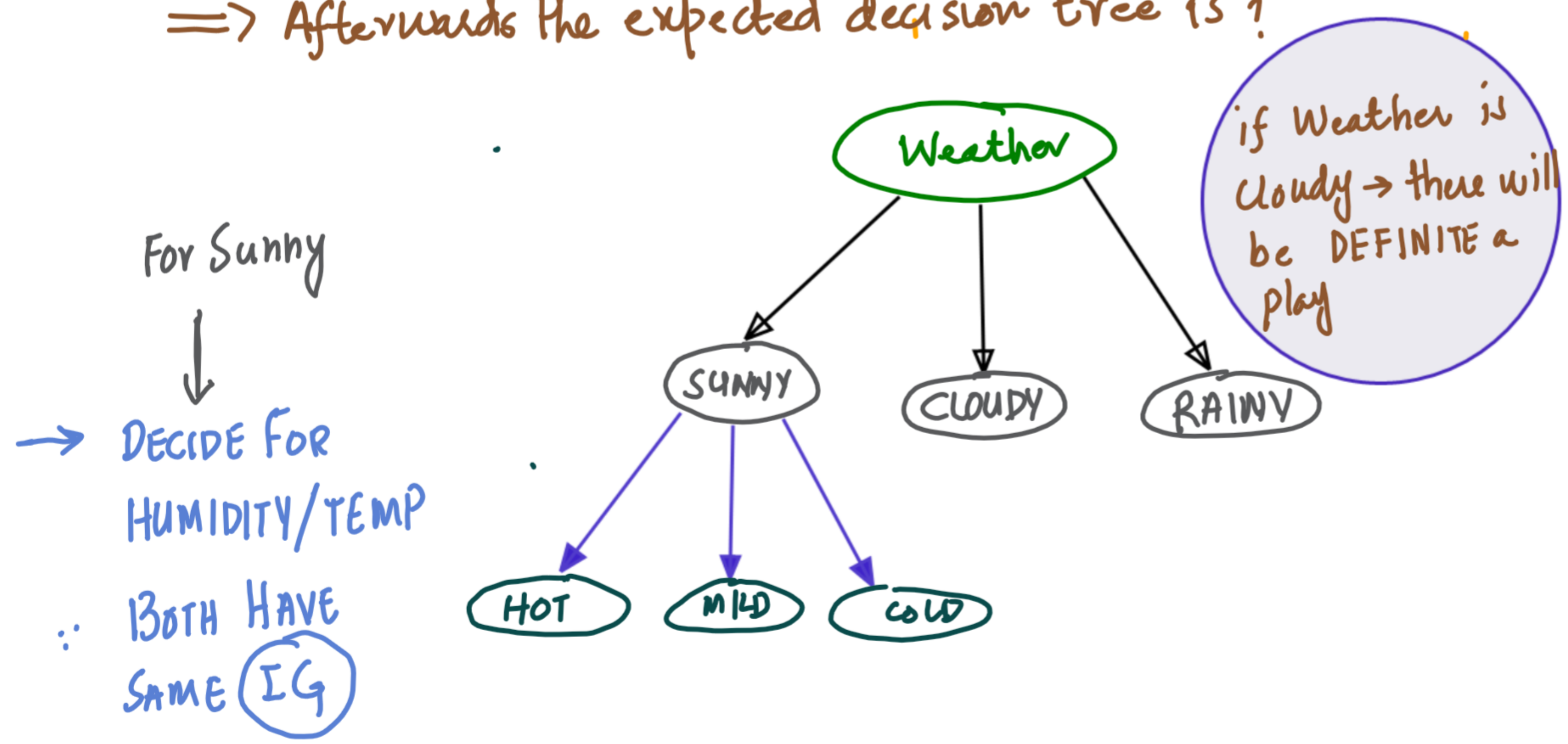
IG = 0.8113 - 0.6887 = 0.1226

IG status:
 Wind = 0.1245
 Humidity = 0.1349
 Weather = 0.4100
 Temperature = 0.2116



→ Solve for this section??

⇒ Afterwards the expected decision tree is ?



For Rainy(???)
 ↓
 → Decide for best feature based on IG?

- I choosed Temp
- Hint: IG of Wind is less than Hum/temp.

PART II

DECISION TREE (FOR REGRESSION PROBLEM) → Also known as REGRESSION TREE

- Predicts a continuous numerical output

• Class (Y/N)

• Reg (Cont.)

- a regression tree asks a series of yes/no questions about the features, splitting the data into increasingly pure subsets. The prediction in each leaf is simply the mean of training targets that fall there.

MATHEMATICAL FOUNDATION:

- SPLITTING CRITERIA → Residual Sum of Square (RSS)

- for best feature / threshold

Partitioning Region (M)

$$RSS = \sum_{m=1}^M \sum_{x_i \in R_m} (y_i - \hat{y}_{R_m})^2$$

The data is divided into 2 regions

feature used for splitting

threshold value

$$R_1(s, t) = \{x | x_s \leq t\}$$

LEFT-Reg

$$R_2(s, t) = \{x | x_s > t\}$$

Right-Reg

The best split minimizes

$$\min_{s, t} \left[\sum_{x_i \in R_1(s, t)} (y_i - c_1)^2 + \sum_{x_i \in R_2(s, t)} (y_i - c_2)^2 \right]$$

Where

c_1 = mean of y_i in R_1

c_2 = " " " " R_2

• Data-Driven Decision

• Data As a Service (DAAS)

EXAMPLE: A restaurant wants to predict daily food demand (Kg) based on number of customers visiting the restaurant. Build a Regression Decision Tree to help the owner in making right choice.

W.HOURS 5 6 8 12 12 15

x_i ?	CUSTOMERS:	20, 30, 40, 50, 60, 70
y_i ?	FOOD DEMAND:	10, 12, 15, 25, 30, 35

SOLUTION:

STEP-I

Calculate Response Mean:

$$\bar{y} = \frac{10+12+\dots+35}{6} = 21.17$$

STEP-II

Calculate RSS:

$$RSS = 530.9$$

$$RSS = \sum (y_i - \bar{y})^2$$

$$RSS = \sum_{i=1}^n (y_i - 21.17)^2$$

$$= (10-21.17)^2 + (12-21.17)^2 + \dots + (35-21.17)^2$$

$$= 124.7 + 84.0 + \dots + 191.4$$

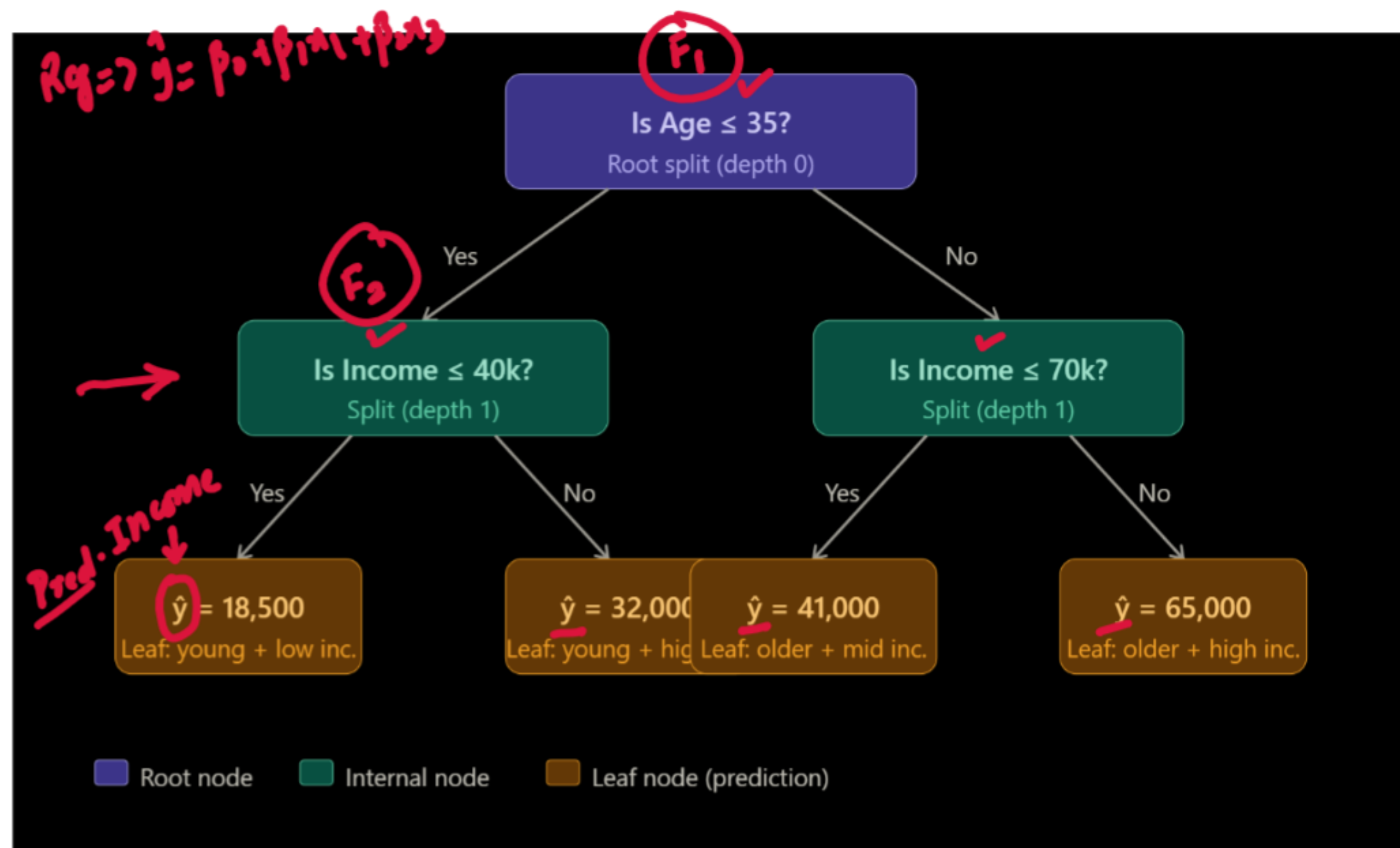
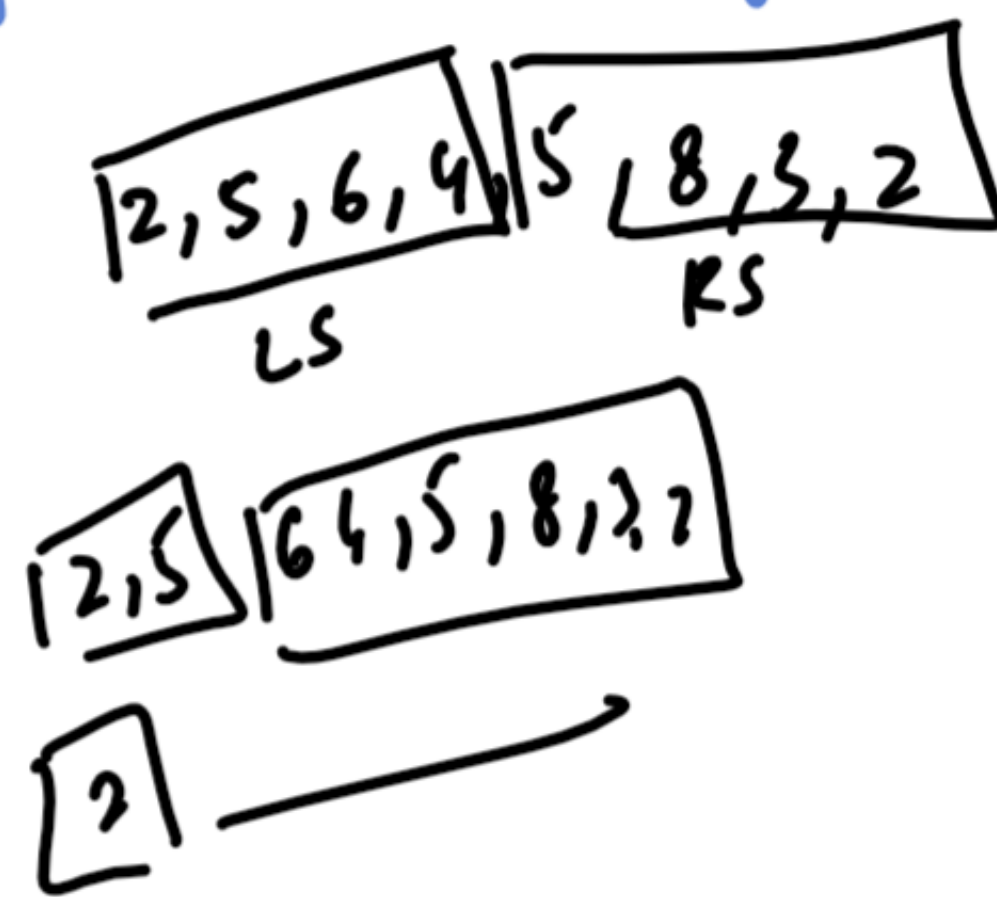


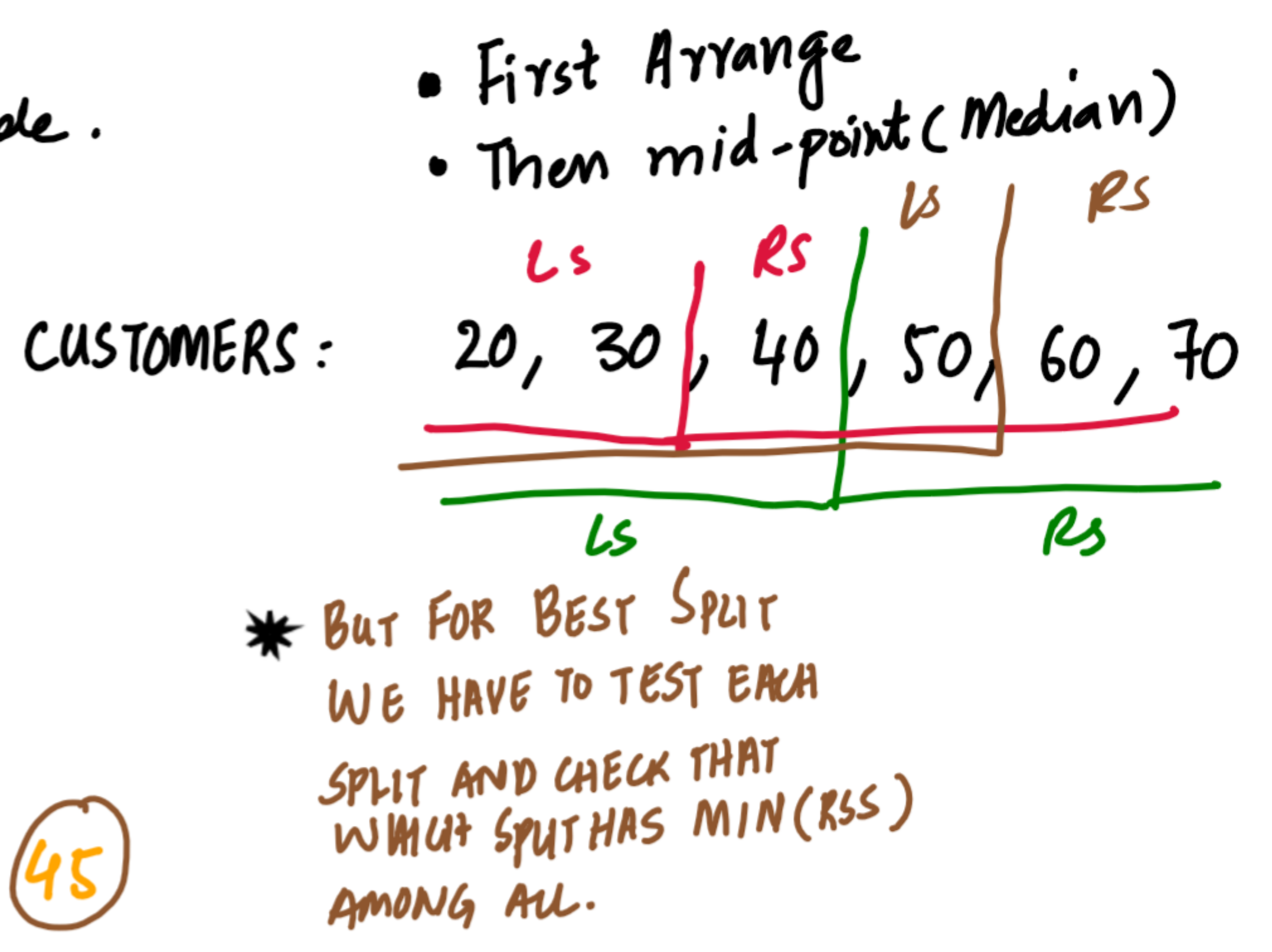
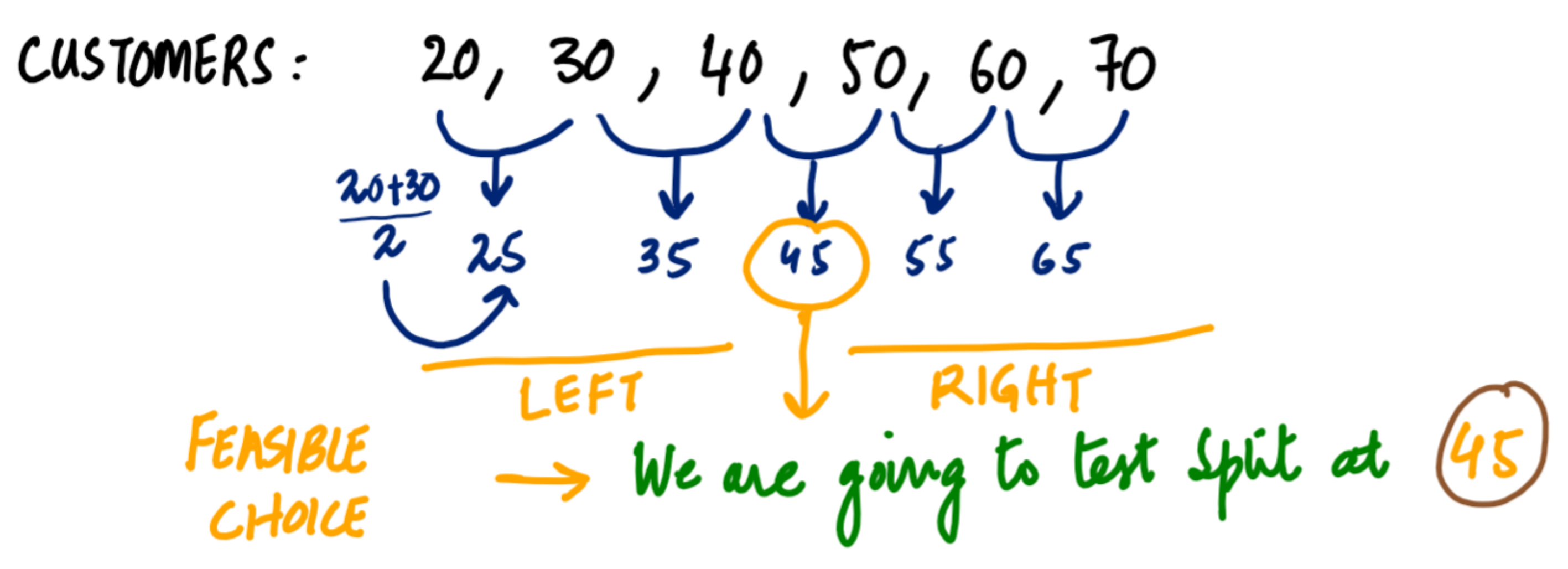
Fig: A regression tree for age vs income expectation.



SELECTION OF
STEP-III(a) Try possible split ← This being based on independent var. (x_i) → predictor

→ One of the choice is mid point from independent variable.

→ Possible split points:



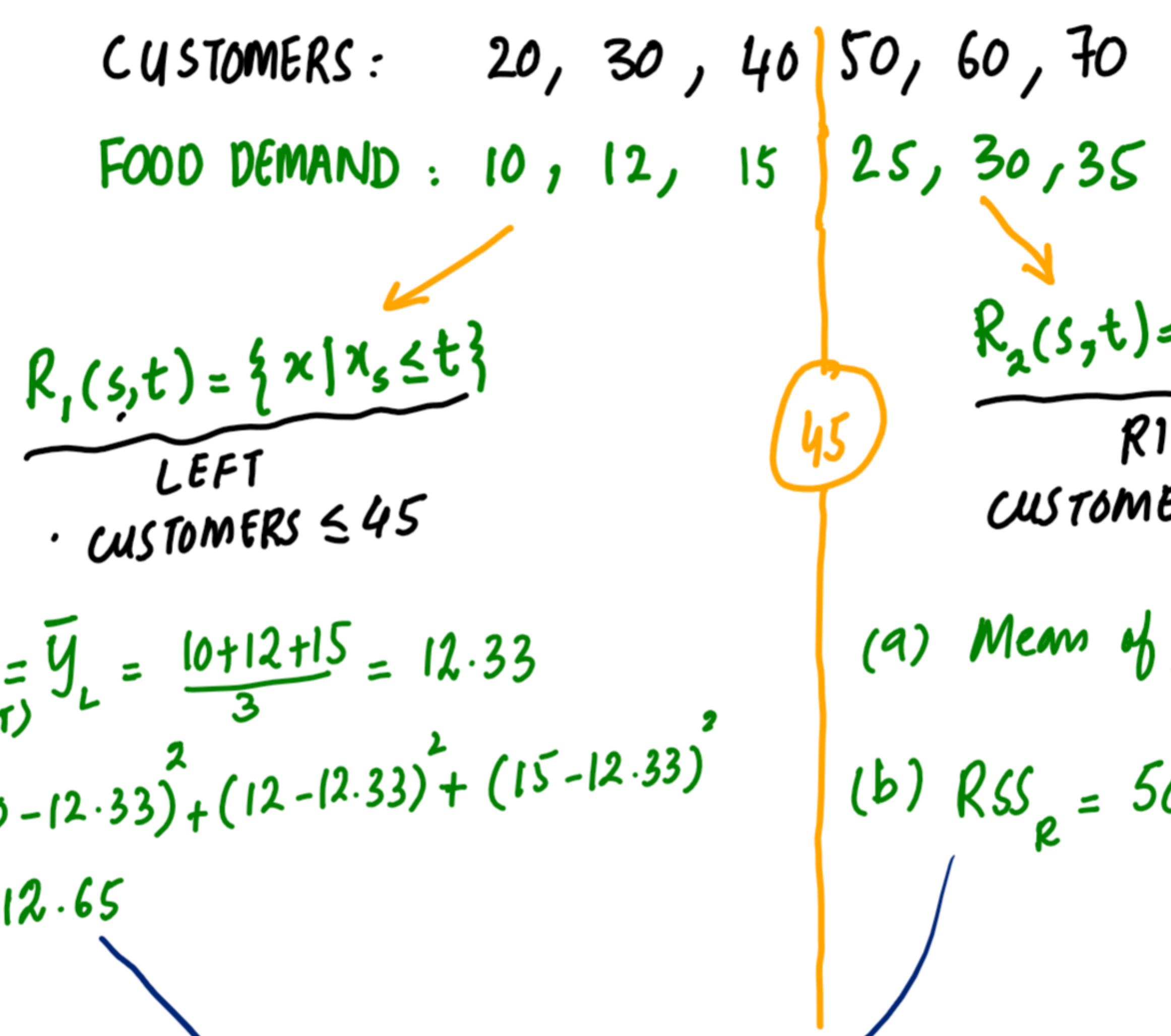
STEP-III(b) Split at 45 Customers:

20, 30 | 40, 50, 60, 70

10, 12 | 15, 25, 30, 35

$\bar{y}_L = 11$
 $RSS_L = (10-11)^2 + (12-11)^2 = 2$

$\bar{y}_R = 26.11$
 $RSS_R = (15-26.11)^2 + (25-26.11)^2 + (30-26.11)^2 + (35-26.11)^2 = 123.11$



STEP-IV Total RSS After Split:

$RSS_{TOTAL} = RSS_L + RSS_R$
 $= 12.65 + 50$
 $RSS_{TOTAL} = 62.65$

** BUT FOR BEST SPLIT WE HAVE TO TEST EACH SPLIT AND CHECK THAT WHICH SPLIT HAS MIN(RSS) AMONG ALL.

Since $RSS_{TOTAL} \ll RSS$, the split significantly reduces error.
 (62.65 $\ll$ 530.9)

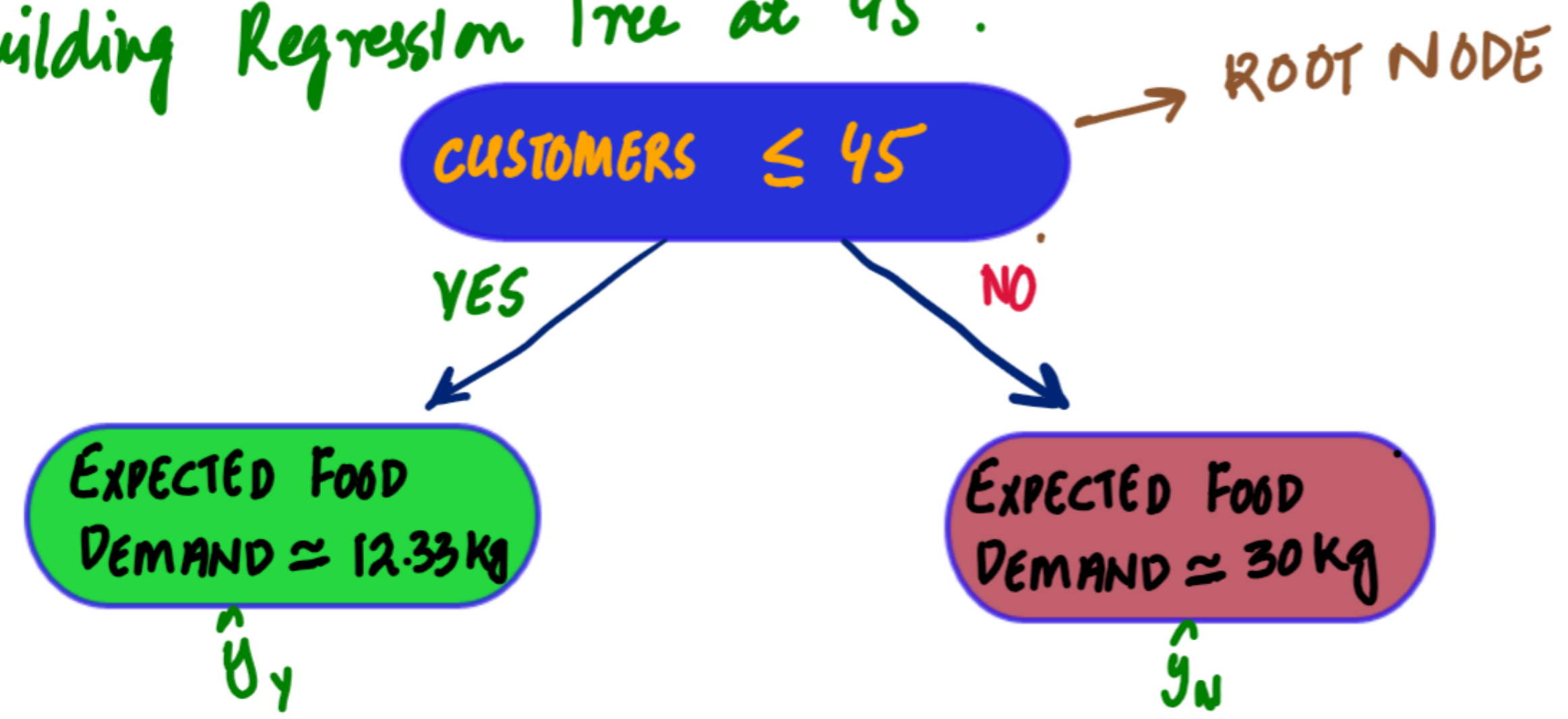
$\min_{s, t} \left[\sum_{x_i \in R_1(s, t)} (y_i - c_1)^2 + \sum_{x_i \in R_2(s, t)} (y_i - c_2)^2 \right]$

: TASK: CHOOSE 35 AND 65 AS SPLIT AND CHECK ITS RSS

STEP-V Building Regression Tree at "45".

• FUNCTION FORM:

$f(x) = \begin{cases} 12.33 & x \leq 45 \\ 30 & x > 45 \end{cases}$



PREDICTION

QUESTION
 Suppose that customer flow on Monday is small (say 15), how much restaurant should expect in food demand?

THIS WILL HELP RESTAURANT IN REDUCING FOOD WASTE + FINANCIALLY

